



Three-dimensional virtual reality-based subjective evaluation of road traffic noise heard in urban high-rise residential buildings

Jin Yong Jeon, Hyun In Jo*

Department of Architectural Engineering, Hanyang University, Seoul 04763, Republic of Korea

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ABSTRACT

In this study, a new noise evaluation method is proposed in which virtual reality (VR) technology in the form of a head-related transfer function (HRTF) and a head-mounted display (HMD) is applied to assess the noise experienced inside residential buildings. Firstly, the sound pressure level and frequency characteristics of road traffic noise recorded in a living room were identified before and after applying the HRTF. Secondly, the subjective response was evaluated in terms of loudness, annoyance, disturbance, and allowance for road traffic noise levels L_{Aeq} of 40–65 dB in four different test environments, namely, without the HRTF or HMD, with the HRTF, with the HMD, and with both the HRTF and HMD. The impacts of the directional information provided by the HRTF and the visual information provided by the HMD were 77% and 23%, respectively. Furthermore, based on an evaluation of the source- and environment-related spatial attributes, the recognition of road traffic noise was the highest with the HRTF, and the recognition of space was the highest with the HMD. When both the HRTF and HMD were applied, the influence of sound externalization in the VR environment increased along with the sense of immersion and realism. Moreover, the recognition of the noise direction and noise width played a significant role in the psycho-acoustic impact, including the annoyance level of the road traffic noise. The findings of this study provide new insights on the influence of external noise on indoor environments.

1. Introduction

In the sound environments of residential buildings, noise sources include nearby neighbors, public spaces such as stairways and corridors, and external roads. In particular, in residential spaces with windows that can be opened in high-density high-rise residential buildings in large cities, road traffic noise is the most common type of external environmental noise that enters the space. The magnitude of the noise is dependent on the ratio of heavy vehicles among the passing vehicles, and thus, it leads to annoyance among the residents and interferes with their activities [1,2]. Consequently, the World Health Organization classifies road traffic noise as a health risk and warns that such noise can cause various problems such as sleep and activity disorders and even heart trouble if the noise level exceeds 55 dBA in the outdoor environment [3,4].

In previous studies, it has been found that the level of annoyance caused by road traffic noise can indicate the impacts of the noise on the health and welfare of the residents [5]. Although the sound pressure level by frequency of road traffic noise is the most significant annoyance factor in most cases, recent studies have shown that residents are influenced not only by the noise levels in their living environments, but

also by environmental factors [6,7]. Previous studies have also shown that variations in the visual elements observed when evaluating environmental noise levels result in variations in noise perception [8,9]. For example, when a photograph of a vegetation-covered soundproof wall was shown, the preference of subjects for the wall increased [10], and when photographs of objects in an urban environment were shown and the response to noise was evaluated, the subjective responses varied depending on whether or not the urban environment was recognized [11,12]. However, the problem with such laboratory evaluations is that photographs or two-dimensional images are shown on screens, which do not accurately reflect the feelings and sensations of actual living environments [10–12]. Although attempts have been made to provide realistic test environments in laboratories through furniture arrangement and interior decoration, there are physical limitations in recreating realistic environments with varying conditions [13].

Virtual reality (VR) technology has advanced to the extent that reality is closely imitated when augmented reality is introduced to realize the visual conditions of the surrounding environment [14] and such technology is being employed for the subjective evaluation of external noise. In these tests, noise sources are evaluated based on their

* Corresponding author. Dept. of Architectural Eng., Hanyang University Seoul 133-791, Republic of Korea.

E-mail address: best2012@hanyang.ac.kr (H.I. Jo).

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positions while providing a sense of realism to test participants in a virtual model space [15,16]. The advantage afforded by VR technology is the ability to simulate various situations within a limited laboratory environment [17,18].

VR technology can be generally classified into stereo sound and virtual space implementation technologies. Three-dimensional (3D) audio technology typically includes sound image localization based on a head-related transfer function (HRTF) or early reflections to position a sound source at a location other than that where the sound source is actually located. This type of audio technology can be further divided into virtual sound field technology for enhancing the sense of the space and sound image externalization technology for moving the sound image to outside the head in headphone reproduction environments [19]. Numerous studies have been performed on various technologies, such as methods of optimizing and calibrating audiovisual environments to ensure accurate auralization in VR. Although it is possible to implement directionality for sound sources by applying such technologies, few studies have been conducted on noise evaluation in such environments [20,21].

In some previous studies, the methods of evaluating actual noise were investigated by implementing a sound environment inside a train carriage via a head-mounted display (HMD), which is a commonly used device for visualizing virtual spaces. However, the evaluation results showed that the annoyance level of the noise inside the train tended to be underestimated [22]. This underestimation was probably due to not only the visual information, but also the realization related to people's expectations, which can influence perception depending on the context (expected or not), that experiencing noise inside a train is inevitable. In recent studies on noise evaluation in VR environments, the interactions between visual, spatial, and directional information in these environments have been investigated. However, as there are many external environments and conditions at work [23,24], few studies have been conducted on interior residential noise.

The purpose of the present study was to apply VR technology to study road traffic noise in multi-story residential buildings, which is the most typical type of external noise in such environments, to investigate the subjective response to the noise. Various noise environments were evaluated depending on whether directional or visual information was provided via the HRTF or HMD, respectively, to assess the subjective response to the road traffic noise entering the building, the noise source itself, and the spatial environment.

2. Methodology

2.1. Measurement

To evaluate the effects of road traffic noise on indoor residents, measurements were taken in 5 min intervals on a weekday afternoon in a standard-size living room in a high-rise residential building that faced a busy eight-lane road. During the testing, a 1/2" microphone (G.R.A.S) and an AS-70 portable sound level meter (RION) were placed at the center of the living room to obtain samples of the road traffic noise and record a single-channel sound source for use in the psycho-acoustic experiment. The measurements were taken at a height of 1.2 m assuming that the listener was seated on the sofa to reflect the normal living conditions of the residents. The road traffic noise was recorded a total of five times, during which the window was either opened or closed. The recorded sound sources showed a sound pressure level distribution in the range of Leq 38.8–55.5 dBA, and the sound pressure level was adjusted using the Amplitude function of the Audition software for auditory evaluation, as shown in Fig. 1. A total of six sound sources were constructed up to Leq 40.0–65.0 dBA. Because sound sources that included reverberations in actual situations were recorded in this study, the reverberation time was neither measured nor corrected.

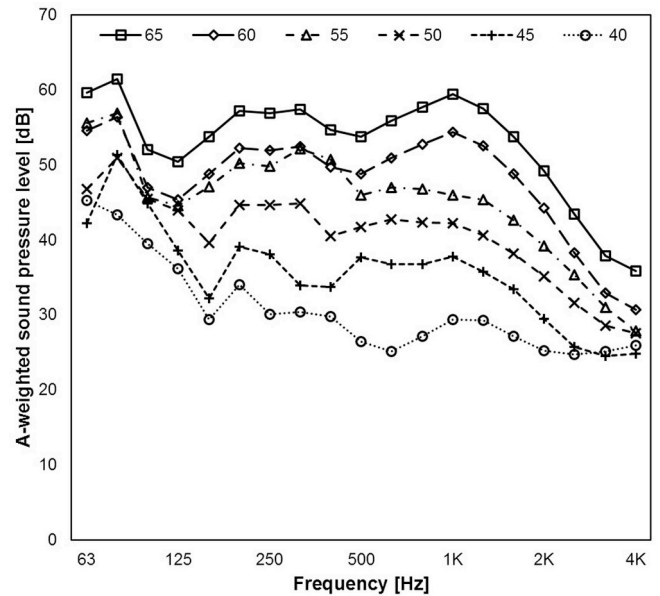


Fig. 1. Frequency characteristics of road traffic noise according to window opening ratio.

2.2. Experimental design

2.2.1. Sound source and virtual room modeling

The HRTF is defined as the impulse response of the sound transmission path when sound is transmitted from a noise source to the eardrum of a test subject [25] and includes the effects of the reflections and diffraction caused by the head, ear, chest, etc. of the subject along the path. Typically, a person can determine the position of a sound source in a 3D space using their two ears. This determination is influenced by the interaural time delay, which affects the time at which the sound reaches each ear; the interaural intensity difference, which is the strength difference recognized between the two ears; and the interaural spectrum difference. The HRTF is an impulse response containing all the information necessary to determine the position of the sound source, and the directionality of the sound source can be realized by convolving the left and right HRTFs with a mono audio signal received through a single channel [17,26,27]. In this study, a CIPIC HRTF database containing the results of measurements of 43 people and two forms of ear on a KEMAR dummy head was employed [28,29]. Ideally, the HRTF should exhibit the best sound image localization performance and sound image externalization effect when applied to the subjects in a study. However, as it is difficult to acquire individual HRTFs due to time constraints, HRTF data were selected and applied by considering the average size, shape, etc. of the human face. The azimuth angle was set to 80°, the elevation angle was set to 0°, and it was assumed that the right-hand window in the living room was open while the resident was facing forward; in this study, the head movement or rotation was not applied. Compared to the original sound, which is depicted in Fig. 1, a difference was detected between the sound pressure levels entering the two ears, while the frequency characteristics were similar, as shown in Fig. 2.

As illustrated in Fig. 3, to investigate the visual impact of the HMD on the traffic noise, modeling was performed using the SketchUp 2018 software package [30] to simulate the living space where the actual sound source was recorded. Furthermore, the Kubity software was used to visualize the model in the HMD. A video was synthesized onto a TV screen to simulate the test subject being seated in the living room watching TV, and the video was muted.

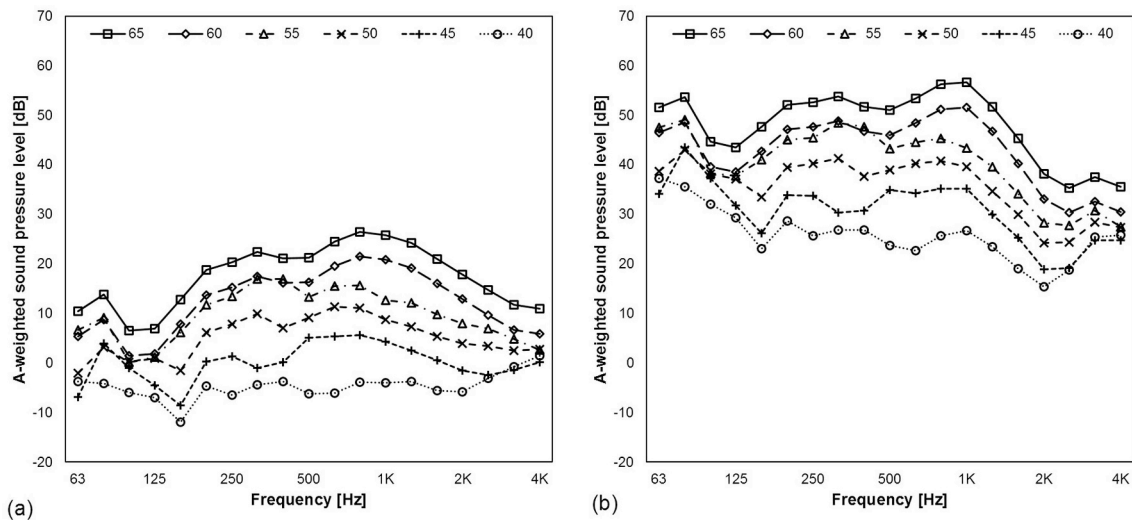


Fig. 2. Frequency characteristics of road traffic noise with the HRTF according to window opening ratio: (a) left channel; (b) right channel.



Fig. 3. 3D virtual room model.

2.2.2. Experimental setup

To investigate the effects of the directional and visual information of the sound source on the road traffic noise entering the building, the noise was evaluated in four different test environments with different HRTF and HMD combinations, as summarized in Table 1. The subjective evaluations were conducted in the semi-anechoic chamber. During this test, the natural background noise of the experimental space was approximately 25 dBA. The measured reverberation time (T30) of the testing space was 0.2 s at the mid-frequency bands. Open-type headphones (Sennheiser HD-650) and an Oculus Rift 2 HMD were used in the evaluation. In total, 12 sound sources were evaluated according to the sound pressure level of the road traffic noise before and after applying the HRTF. The length of each sound source was set to 20 s that is sufficient sound length for subjective evaluation comparing with previous studies [15,31,32], and each was randomly repeated twice to control the response error of the test subject to the initial stimulus. In previous.

2.3. Procedure

A total of 40 test subjects (28 males and 12 females) who had

Table 1
Different experimental setups.

Case	HRTF	HMD
I		
II	○	
III		○
IV	○	○

experienced road traffic noise while living in residential buildings participated in the evaluation. The test subjects were 20–46 years of age (mean age = 26.08, standard deviation = 5.42). To evaluate their hearing prior to the experiment, their auditory thresholds were assessed using an audiometer (Rion AA-77), after which it was determined that all of the subjects had normal hearing.

The test subjects were exposed twice in 2 s intervals to 12 sound sources consisting of six mono-channel sound sources with different noise levels that represented the road traffic noise entering the building, and six sound sources to which the HRTF had been applied. In addition, the subjects completed a questionnaire to capture their magnitude estimates. Each test subject evaluated all four different test environments. To address the physical discomfort of wearing an HMD for a long period of time, the order and break times of the participants were arranged in such a manner that they would not participate consecutively in test environments III and IV.

The survey questions largely fell into three categories: 1) response to the noise, 2) perception of the sound source, and 3) perception of the space. First, the loudness, annoyance, and activity disturbance of the road traffic noise were evaluated on a seven-point numerical scale to examine the subjective responses of the test subjects to the traffic noise. The allowance was defined on a three-point scale: “0: completely acceptable,” “1: somewhat acceptable,” and “2: definitely not acceptable.” The direction and width of the sound source were evaluated on a seven-point numerical scale to determine the differences in perception of the noise source. The quality of immersion, realism, and externalization was evaluated on the same seven-point scale to determine the differences in spatial perception [33].

3. Results

3.1. Subjective response to road traffic noise

The subjective responses of the test subjects to the road traffic noise entering the building were investigated in terms of loudness, annoyance, and activity disturbance, and the results are presented in Fig. 4. As shown in Fig. 4 (a), the perceived noise magnitude increases with increasing sound pressure level and is the highest in the HRTF environment, followed by the mixed HRTF and HMD environment, the NONE environment, and the HMD environment. This trend is similar to that of the annoyance level in Fig. 4 (b), and the correlation with the loudness is quite high (0.72) and statistically significant ($p < 0.01$). The ratings of these two indicators are higher with directional information provided through the HRTF and lower with visual

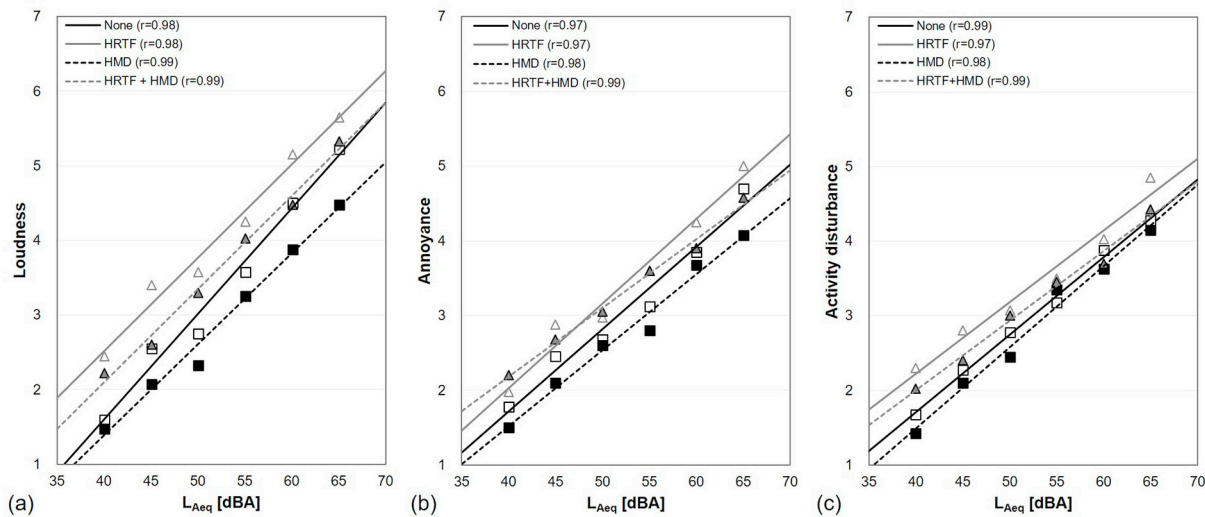


Fig. 4. Mean values of subjective responses as functions of L_{Aeq} for different setups: (a) loudness; (b) annoyance; (c) activity disturbance. (□: None; △: HRTF; ■: HMD; ▲: HRTF + HMD).

information about the space provided via the HMD. The same trend is evident for the activity disturbance in Fig. 4 (c). However, in this case, the differences for each test environment gradually decrease with increasing sound pressure level.

The ratios of the annoyance level and activity disturbance versus L_{Aeq} changes in the four test environments are shown in Fig. 5 (a) and 5 (b). The percentage of annoyance means the ratio of subjects who evaluate the annoyance level > 4 : Moderate. Because the percentage of people who assigned the annoyance level a score of mid-point or higher on the seven-point Likert scale was the same as those who expressed 50% dissatisfaction in a previous study, the same calculation method was applied to the activity disturbance ratings. The percentages of participants who rated each noise level as “2: Definitely not acceptable” upon the occurrence of noise are shown in Fig. 5 (c). Probit regression analysis was conducted to correct the annoyance level and activity disturbance for the evaluation environment. Probit regression analysis involves a type of regression model that can analyze binomial response variables and transform the S-shaped dose response curve to a linear line that can be analyzed by regression through maximum likelihood [34]. All of the probit regression curves were found to be highly correlated with the evaluation data, with correlation coefficients of

0.90 or more.

For the annoyance level, the overall ratio is higher in the environment in which directional information was provided by the HRTF. The difference in the annoyance response due to the HRTF decreased with increasing sound pressure level in the environments in which the HMD did not supply visual information. For activity disturbance, there are clearer differences between groups than for the annoyance level. Furthermore, the visual information effect produced by the HMD is more prominent and the differences between the evaluation environments decrease with increasing sound pressure level of the road traffic noise. For the allowance, the difference in the responses when the HRTF was applied at a low sound pressure level is significant in the environment with the HMD, although the difference gradually decreases with increasing sound pressure level. Conversely, in the environment with no HMD, the impact of the HRTF is insignificant at low sound pressure levels, and it gradually increases with increasing noise level.

When the allowance was set to 50%, which the majority of subjects rated as intolerable, the measurements were approximately 52.5 dBA, 49.5 dBA, 55.4 dBA, and 51.5 dBA in the NONE, HRTF, HMD, and mixed HRTF and HMD environments, respectively.

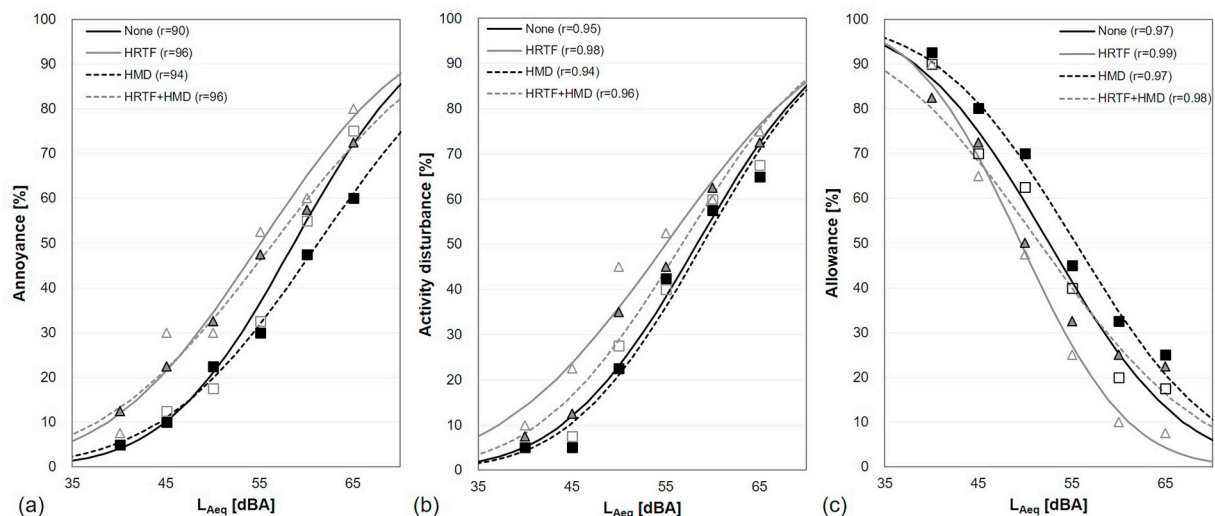


Fig. 5. Percentage of subjective responses as functions of infiltrated road traffic noise level in L_{Aeq} for different setups: (a) annoyance; (b) activity disturbance; (c) allowance. (□: None; △: HRTF; ■: HMD; ▲: HRTF + HMD).

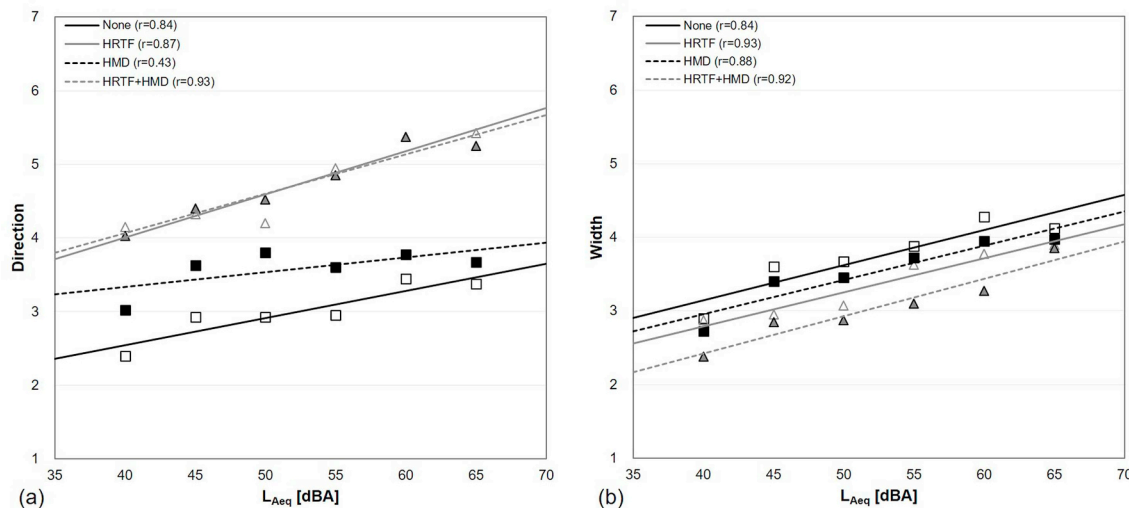


Fig. 6. Source-related spatial sensation as a function of L_{Aeq} for different setups: (a) direction; (b) width. (□: None; △: HRTF; ●: HMD; ◆: HRTF + HMD).

3.2. Source-related spatial attributes

The direction and noise width associated with the recognition of the sound sources in the various environments were evaluated for each level of road traffic noise, and the results are presented in Fig. 6. The direction of the road traffic noise was evaluated higher when HRTF was applied, and it was found that the direction was recognized more when the visual information was given through HMD in the environment in which HRTF was not applied. This is because when the subject evaluates the acoustic-related elements in the HMD environment, the modeled road outside the virtual space is recognized as a noise source, and the sound is heard from that region. When HMDs are provided, the recognition rate of increase with level is reduced by a factor of 0.5 compared to when the HMD is not provided, because it reacts insensitively to the sound change although the direction of the noise source is perceived through the HMD as the dependency to visual information rises. As shown in Fig. 6 (b), the noise width grows increasingly narrow with the HRTF applied and even narrower within the HMD environment. As the sound pressure level increases, the sound width increases and there are no significant differences between the rates of sound width increase in the various environments.

3.3. Environment-related spatial attributes

The difference in recognition in the spatial environments was investigated through immersion, realism, and sound externalization, and the results are shown in Fig. 7. Here, immersion refers to the degree of immersion into the space. Both immersion and realism are the highest in the test environment with the HRTF and HMD simultaneously applied, and both are also higher with the HMD was applied than without it. The increase in the evaluation value with increasing sound pressure level is somewhat reduced with the HRTF applied. In particular, as the noise level increases, the impact of the HRTF decreases and the impacts before and after adding the HMD remain constant despite the increased noise level. This finding suggests that the test subjects were more sensitive to noise level changes in the environment where visual information was provided than in the environment with no augmentation (NONE), and that they were less sensitive to the directional information in environments with high levels of road traffic noise. In the realism evaluation results, the response tendencies versus the evaluation environment are similar, and the differences in responses between the test

environments are more prominent than they were in the immersion tests. The realism is higher in the HMD environment than in the NONE environment and the difference in the HRTF environment is significant. The increase in the environmental recognition evaluation value as a function of the road traffic noise level is smaller in the environment with the HMD than in the environment without it, because the test subjects were more sensitive to the visual information than to the sound information when the road traffic noise was present, as shown in Fig. 6. The sound externalization is less impacted by HMD application than by HRTF application, and the effect is even higher in the environment with directional information about the noise provided. The low externalization values indicate that in-head localization occurred in which the sound appeared to originate from somewhere inside the head, which degraded the feelings of reality and space for the subjects. In contrast, the use of the HRTF resulted in an externalization effect that made the subjects assign high externalization ratings, forming sound images outside their heads. Thus, the above-mentioned problem of in-head localization was partially resolved, similar to the results of a previous study [18]. On the other hand, as shown in Fig. 6 (c), there is almost no change in the externalization response as a function of the sound pressure level.

4. Discussion

4.1. Impacts of audio-visual factors

Two-way ANOVA analysis was conducted to determine whether the effects of the directional information (HRTF) and the visual information (HMD) on the subjective evaluation were statistically significant, and the results are summarized in Table 2. As shown, there is no significant interaction between the HRTF (directional information) and HMD (visual information) and the main effect of each factor is statistically significant ($p < 0.05$). The psycho-acoustic responses of the test subjects, such as the loudness, annoyance, activity disturbance, and allowance, exhibit significant differences from those with HRTF application ($p < 0.01$). In the HRTF environment, the average loudness is 0.73, the annoyance is 0.45, and the activity disturbance is 0.37. The allowance limit was 10% lower for the subjects who evaluated the incoming road traffic noise level as tolerable. Furthermore, the psycho-acoustic responses of the test subjects when using the HMD show significant differences ($p < 0.05$). Consequently, the loudness is 0.43

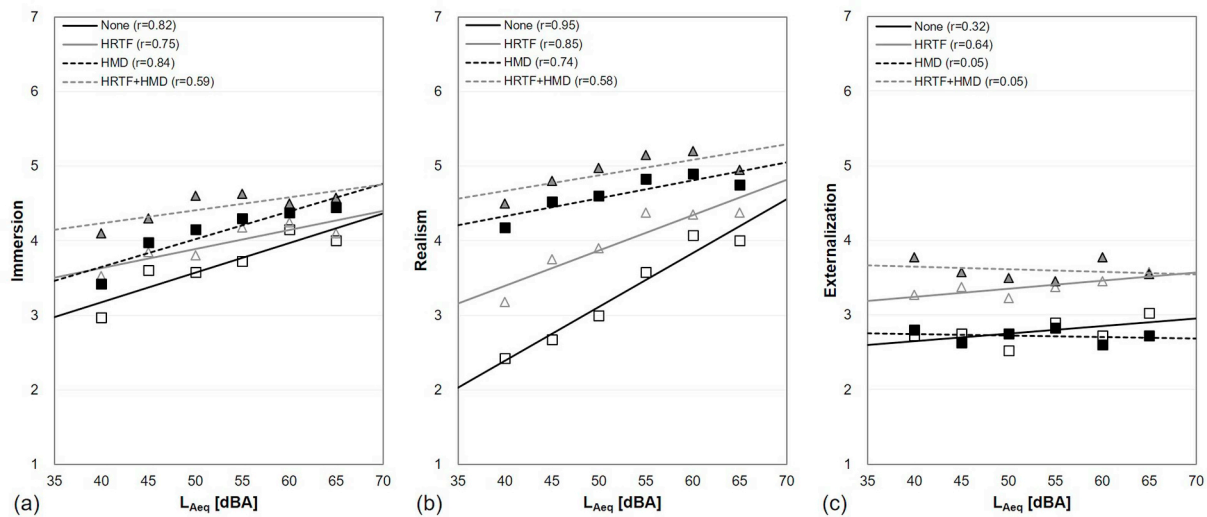


Fig. 7. Environment-related spatial sensation as a function of L_{Aeq} for different setups: (a) immersion; (b) realism; (c) externalization. ($\square$: None; $\triangle$: HRTF; $\circ$: HMD; $\blacktriangle$: HRTF + HMD).

lower on average, the annoyance is 0.21 lower, the activity disturbance is 0.21 lower, and the allowance is 7% higher. The relative contributions of the HRTF and HMD to the subjective responses are 77% and 23%, respectively, on average, which suggests that the test subjects were more sensitive to the HRTF when they evaluated the road traffic noise in the VR environment.

However, because average HRTF data were used for the psychoacoustic tests, there may be individual differences that could result in a larger HRTF contribution. This limitation of the present study will be addressed in future work. In addition, the use of individual HRTFs will enable more precise analysis of the contribution of directional information when evaluating road traffic noise.

4.2. Effects of spatial attributes on subjective responses

The correlation coefficients between the spatial attributes and

subjective evaluation results were calculated and are listed in Table 3, all of which are statistically significant ($p < 0.01$). The loudness, annoyance, and activity disturbance are all positively correlated with the direction, width, immersion, realism, and externalization ($p < 0.01$). Compared to other factors, the noise width has a relatively high correlation and the externalization has a relatively low correlation. In addition, the correlation coefficients between the noise width and the annoyance and disturbance are the highest at 0.52 ($p < 0.01$) and 0.53 ($p < 0.01$), respectively. The allowance limit exhibits negative correlations with all of the spatial attributes, and the correlation coefficient with the noise width is the highest ($r = -0.40$, $p < 0.01$). This result indicates that the sound width determined based on the direction of the sound was one of the factors affecting the comfort of the subjects when they recognized and evaluated the road traffic noise in the living room. Also, the perceptions of the direction, immersion, and realism are statistically significant despite their low magnitudes. Therefore, for

Table 2
Summary of two-way ANOVAs for subjective indicators with HRTF and HMD factors.

Indicator	Factor	Sum of squares	Degree of freedom	Mean square	F	P-value
Loudness	HRTF	127.60	1	127.60	52.98	0.000
	HMD	45.94	1	45.94	19.07	0.000
	HRTF*HMD	0.07	1	0.07	0.03	0.868
	Residual error	2302.38	956	2.41		
	Total	2475.98	959			
Annoyance	HRTF	47.70	1	47.70	18.16	0.000
	HMD	10.42	1	10.42	3.97	0.047
	HRTF*HMD	2.20	1	2.20	0.84	0.360
	Residual error	2511.01	956	2.63		
	Total	2571.33	959			
Activity Disturbance	HRTF	32.27	1	32.27	13.06	0.000
	HMD	10.42	1	10.42	4.22	0.040
	HRTF*HMD	0.60	1	0.60	0.24	0.622
	Residual error	2362.57	956	2.47		
	Total	2405.85	959			
Allowance Limit	HRTF	10.00	1	10.00	16.92	0.000
	HMD	3.04	1	3.04	5.14	0.024
	HRTF*HMD	0.00	1	0.00	0.00	1.000
	Residual error	565.29	956	0.59		
	Total	578.33	959			

Table 3
Correlation coefficients between subjective responses and spatial parameters (**p < 0.01).

Indicator	Direction	Width	Immersion	Realism	Externalization
Loudness	0.37 **	0.46 **	0.36 **	0.32 **	0.21 **
Annoyance	0.35 **	0.52 **	0.38 **	0.34 **	0.22 **
Activity Disturbance	0.33 **	0.53 **	0.36 **	0.31 **	0.20 **
Allowance	−0.29 **	−0.40 **	−0.32 **	−0.29 **	−0.16 **
Limit					

realistic noise evaluation, it is essential that sensory elements for visual and directional implementation are provided in the virtual space using an HRTF and HMD.

The results of the regression analysis performed on the differences between the spatial parameters and the subjective evaluation are presented in Table 4. The overall R^2 is statistically significant, and approximately 30% of the subjective evaluation variables are predicted or explained by the spatial parameters. In terms of the subjective indicators, the width exhibits the highest regression coefficient and the direction coefficient is also significant. Furthermore, although the magnitude of the realism is small, the realism is considered to be partially influential as the annoyance and allowance limit values change by 0.1 standard deviation when the realism value increases by 1 standard deviation.

Two-way ANOVA analysis was conducted to examine the effects of the HRTF and HMD on the evaluation of the spatial attributes, and the results are shown in Table 5. There is no significant interaction between the HRTF and HMD in terms of the width and immersion, and the main effect of each factor is statistically significant ($p < 0.05$).

The recognition of the sound source differs depending on whether or not the HRTF was applied (p & $l_t = 0.001$), and the direction assessments are high and the width low in the HRTF environment. Furthermore, there is a significant difference in the psycho-acoustic response related to the application of the HMD ($p < 0.01$) and the direction assessments are high and the width low in the HMD environment. These results indicate that the HRTF contributed significantly to the recognition of the direction and width of the noise. Moreover, although the HMD had a certain visual impact on the human recognition of the noise, this impact was strongly influenced by the characteristics of the road traffic noise itself, which were enhanced by implementing the VR environment.

There is a statistically significant difference between the cases with and without the HRTF, and the immersion, realism, and externalization results are higher in the HRTF environment. Furthermore, there is a significant difference with the HMD, and the immersion and realism results are higher in the HMD environment. In particular, among the attributes of the spatial environment, the immersion and realism are greatly influenced by the HMD, unlike the sound source recognition results. This finding suggests that humans depend more on visual information in indoor environments than in the changing characteristics of the road traffic noise entering from the outside.

However, because the HRTF and HMD interact with each other, as indicated by the direction and realism evaluation results, the essential

primary effects of each variable were analyzed to characterize them, and the results are shown in Table 6. The statistically significant impact of the HRTF is evident in the direction analysis results, whereas there is a significant difference in the results for the HRTF only without the HMD ($F(1,956) = 21.07$, $p < 0.001$). Thus, there is almost no difference in the direction before and after applying the HRTF with visual information present. The realism demonstrates the significant effect of the HMD regardless of whether the HRTF was applied and conversely shows the significant effect of the HRTF regardless of whether the HMD was applied. Thus, the results suggest that when the HRTF and HMD were applied separately, the feeling of reality changed significantly, and the impact of each factor is statistically significant even with the interaction effect excluded.

4.3. Relationships between the acoustic parameters and spatial attributes

To determine the subjective diffusion and feeling of space of the sound source using the HRTF, the interaural cross-correlation function (IACF) between two sound signals at both ears $f_L(t)$ and $f_R(t)$ was analyzed, which was computed as follows:

$$\Phi_{LR}(\tau) = \lim_{T \rightarrow 0} \frac{1}{2T} \int_{-T}^T f'_L(t) f'_R(t + \tau) dt \quad (1)$$

Where $f'_L(t)$ and $f'_R(t)$ are obtained by signals $f_{L,R}(t)$ after passing through the A-weighted network, which approximately correspond to the ear sensitivity, $s(t)$, so that $f'_{L,R}(t) = f_{L,R}(t) * s(t)$.

As in previous studies, the integration interval ($2T$) was set to 2.5 s and the running step was set to 0.1 s in the analysis [35]. The sound source results obtained with the HRTF applied are shown in Fig. 8. Here, the interaural cross-correlation coefficient (IACC) represents the maximum size of the IACF and has a value between zero and one, which means that the proximity to zero indicates the degree of difference between the sounds on the left- and right-hand sides. The IACC analysis results are presented in Fig. 8 (a). Note that the 65 dBA road traffic noise source used in the experiment was excluded because it had the same frequency characteristics as the 60 dBA noise. After applying the HRTF, the IACC values decrease, and the decrease reaches a maximum of 0.63 at a sound pressure level of 40 dBA. W_{IACC} represents the area of the IACC [36] and is usually related to the apparent source width [37]. As shown in Fig. 8(b), W_{IACC} increases to a maximum of 0.18 at a sound pressure level of 60 dBA.

Table 7 summarizes the correlations between the psycho-acoustic

Table 4
Standardized regression coefficients from multiple linear regression analysis for subjective responses and spatial parameters (*p < 0.05, **p < 0.01).

Indicator	R^2	Direction	Width	Immersion	Realism	Externalization
Loudness	0.30	0.24 **	0.38	−0.01	0.07	0.08 **
Annoyance	0.34	0.19 **	0.44 **	−0.02	0.10 *	0.09 **
Activity Disturbance	0.34	0.19 **	0.47 **	−0.03	0.07	0.08 **
Allowance	0.21	−0.16 **	−0.32 **	−0.01	−0.09 *	−0.05
Limit						

Table 5
Summary of two-way ANOVAs for spatial attributes with HRTF and HMD factors.

Indicator	Set-up	Variation	Sum of squares	Degree of freedom	Mean square	F	P-value
Direction	HRTF	1.14	500.26	1	500.26	261.87	0.000
	HMD	0.29	20.13	1	20.13	10.54	0.001
	HRTF*HMD		20.13	1	20.13	10.54	0.001
Width	HRTF	−0.43	44.20	1	44.20	27.27	0.000
	HMD	−0.26	16.02	1	16.02	9.88	0.002
	HRTF*HMD		0.70	1	0.70	0.43	0.510
Immersion	HRTF	0.31	22.82	1	22.82	12.11	0.001
	HMD	0.47	53.20	1	53.20	28.23	0.000
	HRTF*HMD		0.20	1	0.20	0.11	0.742
Realism	HRTF	0.50	59.50	1	59.50	33.53	0.000
	HMD	1.14	311.68	1	311.68	175.65	0.000
	HRTF*HMD		9.40	1	9.40	5.30	0.022
Externalization	HRTF	0.75	133.07	1	133.07	97.04	0.000
	HMD	0.09	1.80	1	1.80	1.31	0.252
	HRTF*HMD		4.76	1	4.76	3.47	0.063

Table 6
Summary of simple main-effect analysis of data from spatial attribute results.

Indicator	Factor	Sum of squares	Degree of freedom	Mean square	F	P-value
Direction	HMD environ.					
	With	159.85	1	159.85	83.69	0.000
	HRTF					
	Without	360.53	1	360.53	188.76	0.000
	HRTF					
	HRTF environ.					
	With HMD	0.00	1	0.00	0.00	1.000
	Without	40.25	1	40.25	21.07	0.000
Realism	HMD					
	Without	40.25	1	40.25	21.07	0.000
	HMD					
	Error	1826.25	956	1.91		
	HMD environ.					
	With	10.80	1	10.80	6.10	0.014
	HRTF					
	Without	58.10	1	58.10	32.82	0.000
Realism	HRTF					
	HRTF environ.					
	With HMD	106.41	1	106.41	60.12	0.000
	Without	214.67	1	214.67	121.28	0.000
	HMD					
	Without	214.67	1	214.67	121.28	0.000
	HMD					
	Error	1696.34	956	1.77		

test results and physical acoustic parameters. The correlations between the acoustic parameters, including the IACF parameter, and the psycho-acoustic test results are high with the subjective response-related indicators. In particular, the loudness exhibits the highest positive correlation at 0.68. Among the acoustic indicators, the IACC and W_{IACC} exhibit positive correlations with annoyance. This finding is consistent with the results of previous research [38], where the annoyance level of road traffic noise was positively correlated with τ_{IACC} , which represents the variation of the IACC. Furthermore, the IACC and W_{IACC} have the highest positive correlations with the width at 0.32 among the spatial attributes. This result suggests that the IACC and W_{IACC} are somewhat related to noise source width recognition. As shown in Table 4, the noise width could be used as a new measure for evaluating the annoyance levels of humans. However, the correlation between the IACC and direction is low as the IACC exhibits large variations with the sound

pressure level, and the difference is not distinguishable at low sound pressure levels or lower IACC values. The externalization shows no significant correlation with most of the acoustic parameters except IACC, which is related with the directional characteristics. The other noise-related cognitive factors excluding externalization and the space-related cognitive factors show weak correlations with the acoustic parameters. In comparison to them, the correlation coefficients between the subjective response indicators and acoustic parameters are approximately two or more times larger on average. As a result, the subjective response to noise was dominated by the physical parameters related to the sound source level, but the perception of space or sound source showed a generally low correlation with the physical factors, indicating that other factors besides the acoustic characteristics affect the cognitive response.

5. Conclusions

In the present study, the road traffic noise entering a residential building was measured and the frequency characteristics of the noise that entered both ears of subjects was analyzed by synthesizing the HRTF to provide directional information. Furthermore, by modeling the virtual space and providing visual information via an HMD, a subjective psycho-acoustic test was conducted in a VR environment to evaluate various factors, including the subjective, noise recognition, and spatial recognition responses. Based on the test results for different levels of the noise, the impacts of the HRTF and HMD were evaluated together with the factors that impair the entertainment of residents due to the incursion of external road traffic noise into their residential spaces.

The subjective responses to the noise based on the acoustic environment were influenced by the directional and visual information provided by the HRTF and HMD. Because the subjects who participated in this study were aware that they would evaluate traffic noise, they did not reflect the situation in which they did not pay attention to noise while focusing on other tasks in their daily life. However, the tendency of the subjective response to change when the evaluation environment was changed within the same conditions was clear. The contribution of the HRTF to subjective response was 77%, which was higher than that of the HMD at 23%. Thus, the results suggest that the test subjects were susceptible to the directional information associated with the noise. When the factors related to the recognition of the noise and the source- and environment-related spatial attributes were evaluated, they showed high correlations with the noise width (0.48 on average), which further emphasizes the correlations with the direction, immersion, and realism.

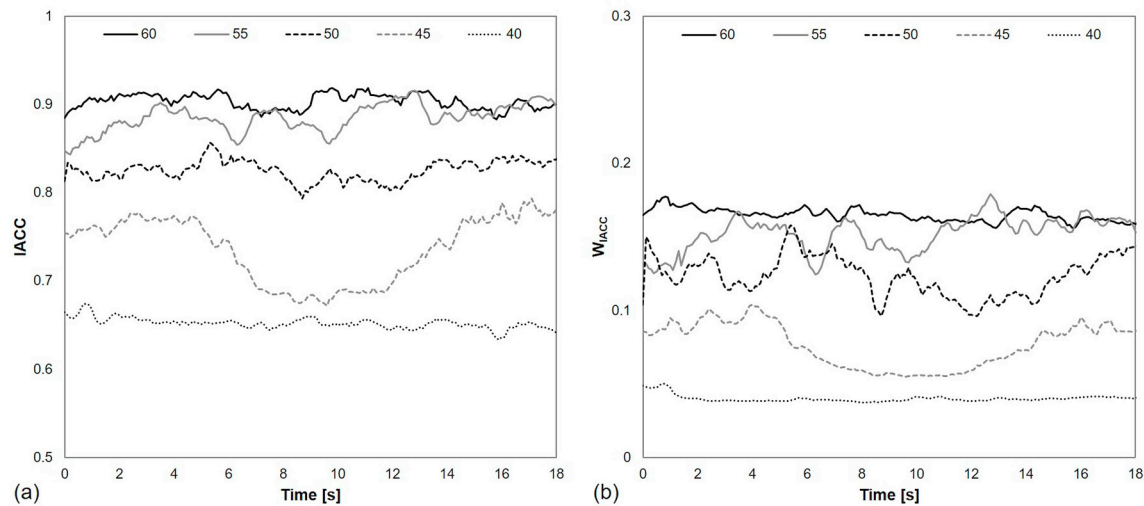


Fig. 8. Interaural cross-correlation of road traffic noise: (a) IACC; (b) W_{IACC} .

Table 7

Correlation coefficients between subjective evaluation indicators and acoustic parameters (bold means $p < 0.01$).

Indicator	L_{Aeq}	L_{Amax}	L_{A90}	L_{A50}	L_{A10}	$L_{A10}-L_{A90}$	IACC	W_{IACC}
Loudness	0.68	0.68	0.68	0.68	0.68	-0.12	0.07	0.66
Annoyance	0.54	0.55	0.54	0.55	0.54	-0.09	0.10	0.52
Disturbance	0.54	0.54	0.54	0.54	0.54	-0.09	0.10	0.52
Allowance	-0.52	-0.52	-0.52	-0.52	-0.52	0.11	-0.12	-0.52
Direction	0.23	0.23	0.23	0.23	0.23	-0.02	-0.25	0.32
Width	0.32	0.32	0.31	0.32	0.32	-0.01	0.26	0.27
Immersion	0.18	0.18	0.17	0.18	0.19	0.03	-0.03	0.21
Realism	0.24	0.24	0.23	0.24	0.24	0.02	-0.05	0.28
Externalization	0.03	0.03	0.03	0.03	0.03	-0.03	-0.24	0.08

The recognition of the noise was influenced by the directional information provided via the HRTF. In particular, the impact on the recognition of the direction significantly increased. The recognition of the space was influenced by the use of the HMD, and the impact on the recognition of the realism significantly increased. These findings are consistent with the results of audio-visual evaluation studies that were conducted in urban sound environments [15,16].

The results of this study confirm the applicability and necessity of an HRTF and HMD when evaluating noise in residential buildings. Furthermore, if additional head rotation and room impulse responses

are included in the evaluation, this approach can be effectively utilized for noise evaluation by implementing identical realistic environments in the limited conditions of laboratories.

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Appendix A. Questionnaire for subjective evaluation (Translated from Korean)

1. How loud is the presented road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely
☐ ☐ ☐ ☐ ☐ ☐ ☐

2. How annoying is the presented road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely
☐ ☐ ☐ ☐ ☐ ☐ ☐

3. How disturbing is the presented road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely
☐ ☐ ☐ ☐ ☐ ☐ ☐

4. Would the sound be acceptable if it could be heard in your own home?

0: Definitely not acceptable 1: Acceptable to some extent 2: Completely acceptable
☐ ☐ ☐

5. How distant is the presented road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely
☐ ☐ ☐ ☐ ☐ ☐ ☐

6. How directive is the presented road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely
☐ ☐ ☐ ☐ ☐ ☐ ☐

7. How large is the perceived width of the road traffic noise?

1: Not at all 2: Very little 3: A little 4: Moderately 5: Fairly 6: Very 7: Extremely

☐	☐	☐	☐	☐	☐	☐
8: How immersive is the presented evaluation environment?						
1: Not at all	2: Very little	3: A little	4: Moderately	5: Fairly	6: Very	7: Extremely
☐	☐	☐	☐	☐	☐	☐
9: How realistic is the presented evaluation environment?						
1: Not at all	2: Very little	3: A little	4: Moderately	5: Fairly	6: Very	7: Extremely
☐	☐	☐	☐	☐	☐	☐
10: How much is the perceived externalization of the road traffic noise?						
1: Not at all	2: Very little	3: A little	4: Moderately	5: Fairly	6: Very	7: Extremely
☐	☐	☐	☐	☐	☐	☐

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